

LAPORAN PRAKTIKUM

TEKNOLOGI Iot *Lab 03*

DOSEN :

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Kelas : TRI 3B

PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

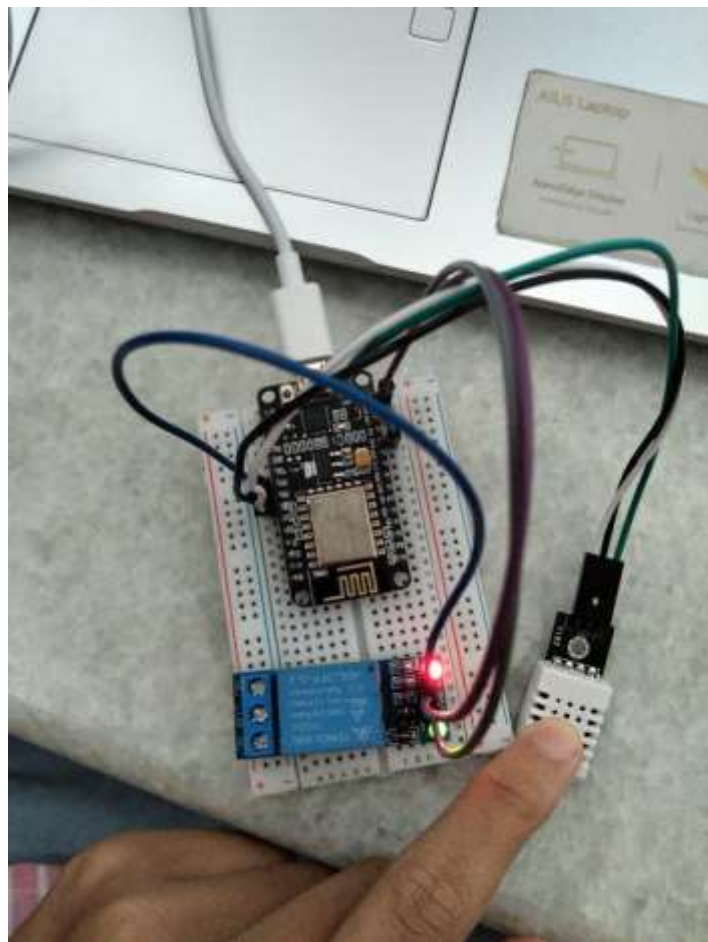
2026

client

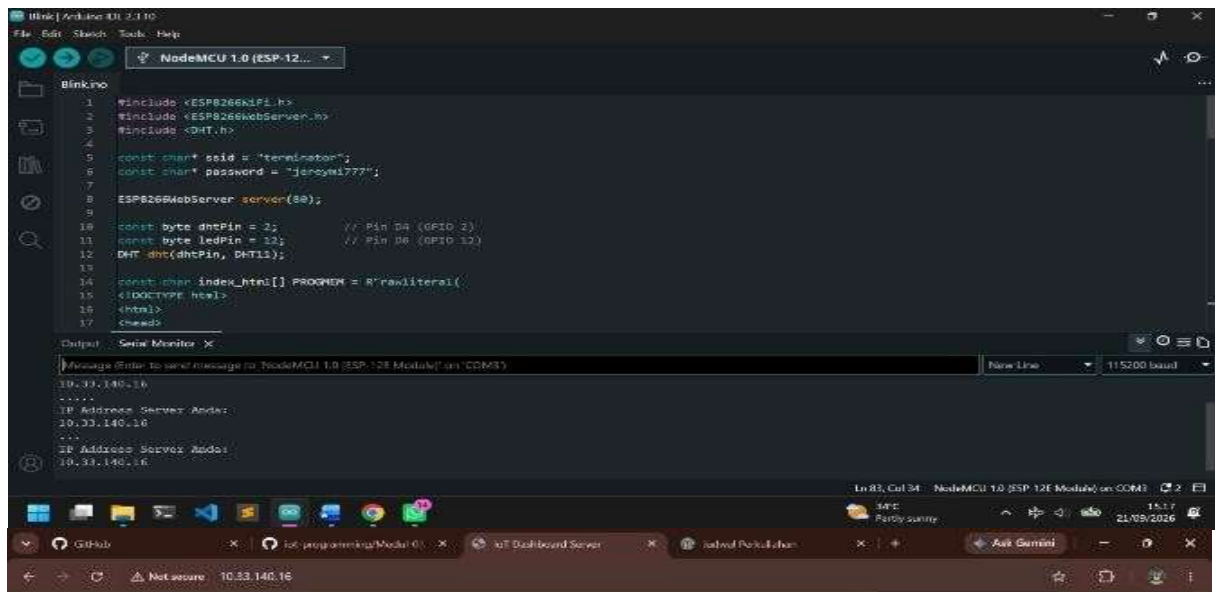
```

1 #include <SPI.h>
2 #include <SoftwareSerial.h>
3 #include <WiFi.h>
4
5 const char* ssid = "client";
6 const char* password = "password123";
7
8 // Define IP and server target
9 const char* serverName = "http://192.168.1.100/relay/";
10 const byte port = 80;
11
12 void setup() {
13   Serial.begin(115200);
14   WiFi.begin(ssid, password);
15   while (WiFi.status() != WL_CONNECTED) {
16     delay(1000);
17     Serial.print(".");
18   }
19   Serial.println("WiFi connected");
20 }
21
22 void loop() {
23   // Send data to server
24   // For example: GET / HTTP/1.1
25   // GET / HTTP/1.1
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98   // GET / HTTP/1.1
99   // GET / HTTP/1.1
100  // GET / HTTP/1.1

```



Server



The screenshot shows the Arduino IDE interface. The code in the main editor is as follows:

```
1 #include <ESP8266WiFi.h>
2 #include <ESP8266WebServer.h>
3 #include <DHT.h>
4
5 const char* ssid = "terminator";
6 const char* password = "j0rg3m1777";
7
8 ESP8266WebServer server(80);
9
10 const byte dhtPin = 2; // Pin D4 (GPIO 2)
11 const byte ledPin = 12; // Pin D6 (GPIO 12)
12 DHT dht(dhtPin, DHT11);
13
14 const char index_html[] PROGMEM = R"rawliteral(
15 <!DOCTYPE html>
16 <html>
17 <head>
```

The Serial Monitor at the bottom shows the following output:

```
10.33.140.16
....
IP Address Server Addr:
10.33.140.16
...
IP Address Server Addr:
10.33.140.16
```

ESP8266 Web Server

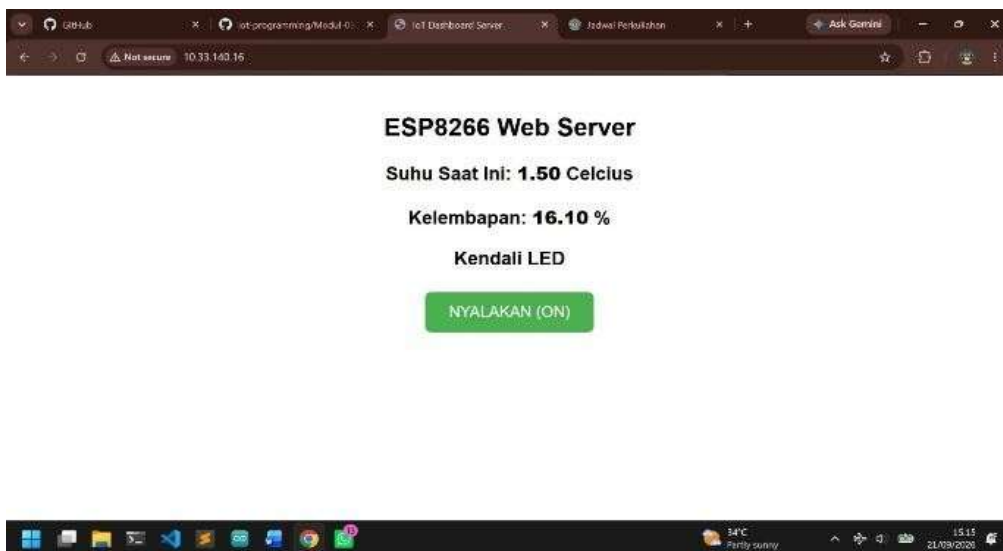
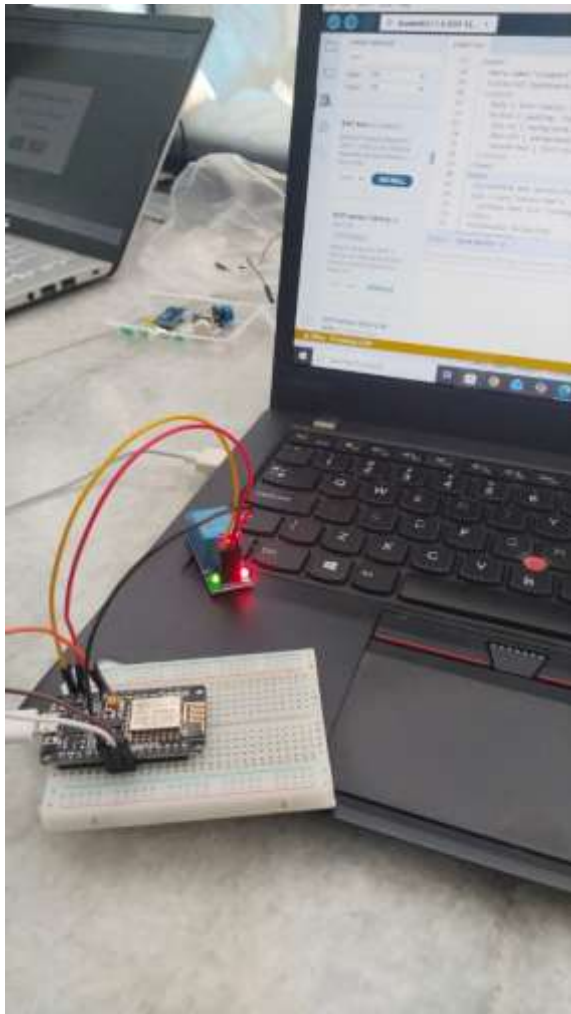
Suhu Saat Ini: **1.50** Celcius

Kelembapan: **16.10** %

Kendali LED

MATIKAN (OFF)





Berdasarkan hasil pengujian sistem komunikasi *Machine-to-Machine* (M2M), NodeMCU

Server dengan alamat IP 10.33.140.16 berhasil menyajikan data telemetri suhu (1.50 °C) dan kelembapan (16.10 %) secara *real-time*. Di sisi lain, NodeMCU *Client* secara otomatis membaca nilai analog LDR. Saat nilai pembacaan melebihi batas yang ditentukan (*ldrValue* > 300), *Client* mengirimkan permintaan *HTTP GET* yang menghasilkan kode respons 303. Respon ini memicu pin D6 (GPIO 12) pada *Server* menjadi *HIGH*, sehingga LED indikator secara fisik menyala. Tampilan pada web *dashboard* pun berubah menjadi tombol merah "MATIKAN (OFF)". Pengujian ini membuktikan bahwa koordinasi M2M antar mikrokontroler berjalan secara stabil melalui jaringan nirkabel lokal.